

Exercise 11: Frank-Wolfe Algorithm

*Lecturer: Aurelien Lucchi***Problem 1 (Affine invariance property of Frank-Wolfe):**

1. An important property of Frank-Wolfe is that its updates are affine invariant. Demonstrate why this property holds.
2. How does the curvature constant M' of h compare to the curvature constant M of f ?

Problem 2 (Induction for Frank-Wolfe analysis):

In the lecture, we have seen that

$$h(\mathbf{x}_k) \leq (1 - \gamma_k) h(\mathbf{x}_{k-1}) + \gamma_k^2 C,$$

where $C = \frac{M}{2}$ and $\gamma_k = \frac{2}{k+2}$. Prove that

$$h(\mathbf{x}_k) \leq \frac{4C}{k+2}.$$

Problem 3 (Curvature constant):

Recall the definition of the curvature constant of f described in the lecture notes:

$$M := \max_{\substack{\mathbf{x}, \mathbf{s}, \mathbf{y} \\ \mathbf{y} = (1-\gamma)\mathbf{x} + \gamma\mathbf{s}}} \frac{2}{\gamma^2} (f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})). \quad (1)$$

If the function f has L -Lipschitz gradients, prove that the curvature constant satisfies $M \leq \text{diam}^2(\mathcal{D}) \cdot L$, where

$$\text{diam}(\mathcal{D}) = \max_{\mathbf{x}, \mathbf{s} \in \mathcal{D}} \|\mathbf{x} - \mathbf{s}\|_2. \quad (2)$$

Problem 4 (Inexact updates):

Assume we don't exactly solve the subproblem at each step but instead require

$$\nabla f(\mathbf{x}_{k-1})^\top \mathbf{s}_{k-1} \leq \min_{\mathbf{s} \in C} \nabla f(\mathbf{x}_{k-1})^\top \mathbf{s} + \frac{M\gamma_k}{2} \cdot \delta, \quad (3)$$

where δ is some inaccuracy parameter. Adapt the analysis of Frank-Wolfe provided in the lecture to show that the following rate holds:

$$f(\mathbf{x}_k) - f^* \leq \frac{2M}{k+2} (1 + \delta). \quad (4)$$

Problem 5 (Programming):

complete TODOs in the Jupyter Notebook provided.